PAPER_537 — Session 140 Hub: grok_share_0f5d4c91f2c.txt — DPM Shell-Energy Radiance, Negative Time, and DPM-Unified Forces

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER_537_Session140_Hub_DPM_Shell_Radiance_Negative_Time_Forces.md

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Author: Daniel T. Murphy **Framework:** Star-Magic / UQFF **Version:** v5.00 **Date:** 2026-03-25 **Session:** 140 — grok_share_0f5d4c91f2c.txt **CP4 Class:** Session140GrokShare0f5d4c91f2cHubCalculator (#115)

§1 — Session Overview

Source document: `grok_share_0f5d4c91f2c.txt` **Origin:** BigBangHypergraphTheory_12Dec2025.docx follow-up recalculation **Position in pipeline:** Continuation of Session 136 (BigBangHypergraph); recalculation incorporating DPM correction, negative time proof, and force unification.

Papers generated: PAPER_533–537 (5 papers) **CP4 classes introduced:** #111–#115 (5 classes + this hub)

§2 — Corrections to Session 136

Session 136 encoded the `BigBangHypergraphTheory_12Dec2025.docx` content (fully integrated into the codebase). Session 140 introduces the following **corrections and refinements** from the Grok recalculation follow-up:

#	Item	Prior Form	Session 140 Upgrade		
1	DPM encapsulation	"SCm encapsulates"	DPM reaction forms layered shell-energies		
2	Phase cascade	Unordered	quantum-multi-fields→plasma→gas→liquid→solid		
3	t_{adj}	$t_{\text{obs}}/(1+\Delta_{\text{rel}})$	$t_{\text{obs}}/(1+\Delta_{\text{dil}}) + t_{\text{neg}}$		
4	Spooky distance	Qualitative only	$Distance_{\text{spooky}} = c \cdot	t_{\text{neg}}	$
5	Dual existence	Not defined	$DualExist = \int_{t_{\text{pos}}}^{t_{\text{neg}}} Existence\, dt$		
6	F_{inert}	Not a pure force	$-\partial(DPM_{\text{react}} \cdot SE)/\partial v^{26} \cdot t_{\text{neg}}$		
7	F_{centrip}	$m \omega^2 r$ (classical)	$DPM_n \cdot \omega_{CW}^2 \cdot r^l / (1+\Delta_{\text{dil}})$		
8	F_{centrif}	Fictitious (classical)	$DPM_s \cdot \omega_{CCW}^2 \cdot r^l \cdot t_{\text{neg}}$ (pure)		

#	Item	Prior Form	Session 140 Upgrade		
9	$Prob_{\text{order}}$	$(v_i - v_c)$ factor only	$\times (1 + \Delta_{\text{dil}} \cdot t_{\text{neg}})$		

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§3 — New Physics Assets (Session 140)

PAPER_533 — DPM Layered Shell-Energy Radiance Phase Cascade

CP4 #111 — DPMLayeredShellEnergyRadianceCalculator

$$ShellEnergy^{(l)} = \int Radiance_{\text{quant}}\, dt_{\text{neg}}$$ $$DPM_{\text{react}} = \frac{\kappa(DPM_n - DPM_s)}{r^{26}} + \frac{\partial^{26} Grind_{\text{opp}}}{\partial t^{26}_{\text{adj}}}$$

Triple-calc: Layer 1 (CW), Layer 2 (CCW), Layer 3 (t_{neg}). Phase cascade: quantum-multi-fields → plasma → gas → liquid → solid.

PAPER_534 — Negative Time Dilation Proof

CP4 #112 — NegativeTimeDilationSpookyDistanceCalculator

$$t_{\text{adj}} = \frac{t_{\text{obs}}}{1+\Delta_{\text{dil}}} + t_{\text{neg}}$$ $$Distance_{\text{spooky}} = c \cdot |t_{\text{neg}}|$$ $$DualExist = \int_{t_{\text{pos}}}^{t_{\text{neg}}} Existence\, dt$$

Observable $\Delta_{\text{dil}} \neq 0$ proves $t_{\text{neg}} < 0$.

PAPER_535 — DPM-Unified Forces

CP4 #113 — DPMUnifiedInertiaCentripetCentrifugCalculator

$$F_{\text{inert}} = -\frac{\partial(DPM_{\text{react}} \cdot ShellEnergy)}{\partial v^{26}} \cdot t_{\text{neg}}$$ $$F_{\text{centrip}} = \frac{DPM_n(SCm) \cdot \omega_{CW}^2 \cdot r^l}{1+\Delta_{\text{dil}}}$$ $$F_{\text{centrif}} = DPM_s(UA') \cdot \omega_{CCW}^2 \cdot r^l \cdot t_{\text{neg}}$$ $$F_{\text{inert}} = F_{\text{centrip}} - F_{\text{centrif}} \quad [M = F_{\text{inert}}/a^{26}]$$

Resolves classical fictitious-force conundrum: all 3 are pure DPM-emergent.

PAPER_536 — Shell Radiance Prototype Equation

CP4 #114 — ShellRadiancePrototypeEquationCalculator

Full assembled system: ProtoH, U_b, BigBang trigger, $Prob_{\text{order}}$ with $(1+\Delta_{\text{dil}} \cdot t_{\text{neg}})$ factor. Master form:

$$\Psi_{26D}(t_{\text{adj}}) = ProtoH + U_b \cdot Prob_{\text{order}} + BigBang \cdot \exp\!\left(-\frac{|t_{\text{neg}}|}{t_{\text{adj}}}\right)$$

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§4 — CP4 Registry Update

Class	#	Paper	Status
DPMLayeredShellEnergyRadianceCalculator	111	PAPER_533	Implemented
NegativeTimeDilationSpookyDistanceCalculator	112	PAPER_534	Implemented
DPMUnifiedInertiaCentripetCentrifugCalculator	113	PAPER_535	Implemented
ShellRadiancePrototypeEquationCalculator	114	PAPER_536	Implemented
Session140GrokShare0f5d4c91f2cHubCalculator	115	PAPER_537	Implemented

CP4 total classes: 108 (103 prior implementations + 5 Session 140) **CP4 `__all__` entries:** 115 (110 prior + 5 Session 140)

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§5 — OutputData Registration

`SOURCE180_SESSION140_RESULTS` (document_id=25) registered in `CondensedPhysics_OutputData.py` with complete equation set, 8 new physics items, mass equilibrium, triple-calc system, canonical constants, and 5/5 validation tests passed.

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§6 — Integration Confirmation

All Session 140 physics verified **not present** in pre-existing codebase:

- No `DualExist` math (prior: `QuantumEntanglementTerm` qualitative only)

- No `Distance_spooky = c·|t_neg|` (prior: qualitative spooky reference only)
- No DPM-based F_{inert}/F_{centrip}/F_{centrif} (prior: classical $m\omega^2 r$ form in `compute_centripetal_centrifugal()`)
- No t_{adj} with $+t_{\text{neg}}$ term (prior: missing that term)
- No $(1+\Delta_{\text{dil}} \cdot t_{\text{neg}})$ factor in $Prob_{\text{order}}$
- No ordered phase cascade (prior: unordered)

Session 140 integration is additive and backward-compatible.

CP4 v5.00 — Session 140 complete.